AD154 Datasheet

Zhuhai Jieli Technology Co.,LTD

Version：V1.2

Date：2024.11.20

ZHUHAI JIELI TECHNOLOGY CO.,LTD

Copyright © Zhuhai Jieli Technology Co.,LTD. All rights reserved.

AD154 Features

CPU Core

- 32-bit CPU,Built-in ICACH, can be connected to Flash for expansion of code
- The main frequency is up to 120MHz

Memory

- On-chip SRAM 20kbyte
- Off-chip Flash memory is required

Clock Source

- RC Clock frequency about 16MHz
- LRC(low power RC) clock frequency about 200KHz

Digital I/O

- Up to 28 programmable digital I/O pins
- General the IO supports pull-up(10k),pull-down(60k), strong,weak output,input and high impedance
- Up to 12 external interrupt/wake-up source(low power available,can be multiplexed to any I/O, with hardware filter)
- Input channel and Output channel, provide arbitrary IO input and output options for some modules

Digital peripherals

- Two UART Controllers(UART0/1) supports DMA and Flow Control
- Two SPI Controllers with DMA(SPI0/1) support master mode and slave mode,SPI0 support 4bit,SPI1support 2bit
- One Spi Flash Controller to run code
- One SD host controller
- Three 32-bit Asynchronous Divider Timers
- One IIC Controller
- Four channel PWM output
- Infrared remote control decoder
- Watchdog

Analog Peripherals

- 0.5 watt Class-D audio amplifier output
- 14 channel 10-bit high precision ADC
- Low voltage protection
- Power on reset

PMU

- Soft off current: <2uA
- Power down current: <30uA
- Music mode: <6mA
- LVD range(3bit):1.8V~2.5V, step100mV
- VBAT range : 2.0V~5.5V
- HPVDD range : 2.0V~5.5V
- VDDIO range : 2.0V~3.4V

Temperature

- Operating temperature: -40℃ to +85℃
- Storage temperature: -65℃ to +150℃

Package

- LQFP48(7mm*7mm)

Application

- Sound Toy
- Audio player

ZHUHAI JIELI TECHNOLOGY CO，LTD

Confidential
The information contained herein is the exclusive property of JIELI and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of JIELI.

1 Block Diagram

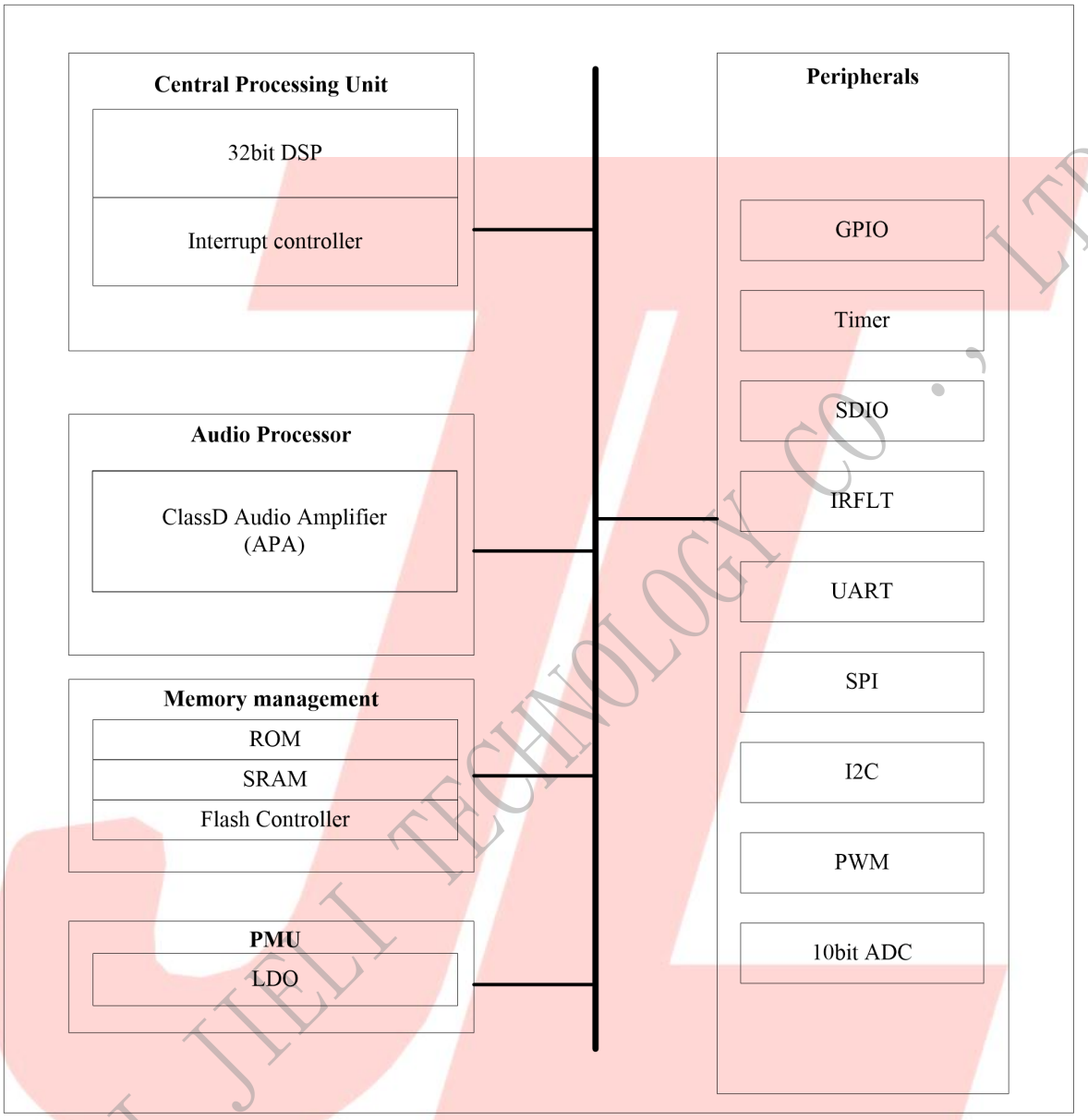


Figure 1-1 AD154A Block Diagram

ZHUHAI JIELI TECHNOLOGY CO., LTD

Confidential
The information contained herein is the exclusive property of JIELI and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of JIELI.

2 Pin Definition

2.1 Pin Assignment

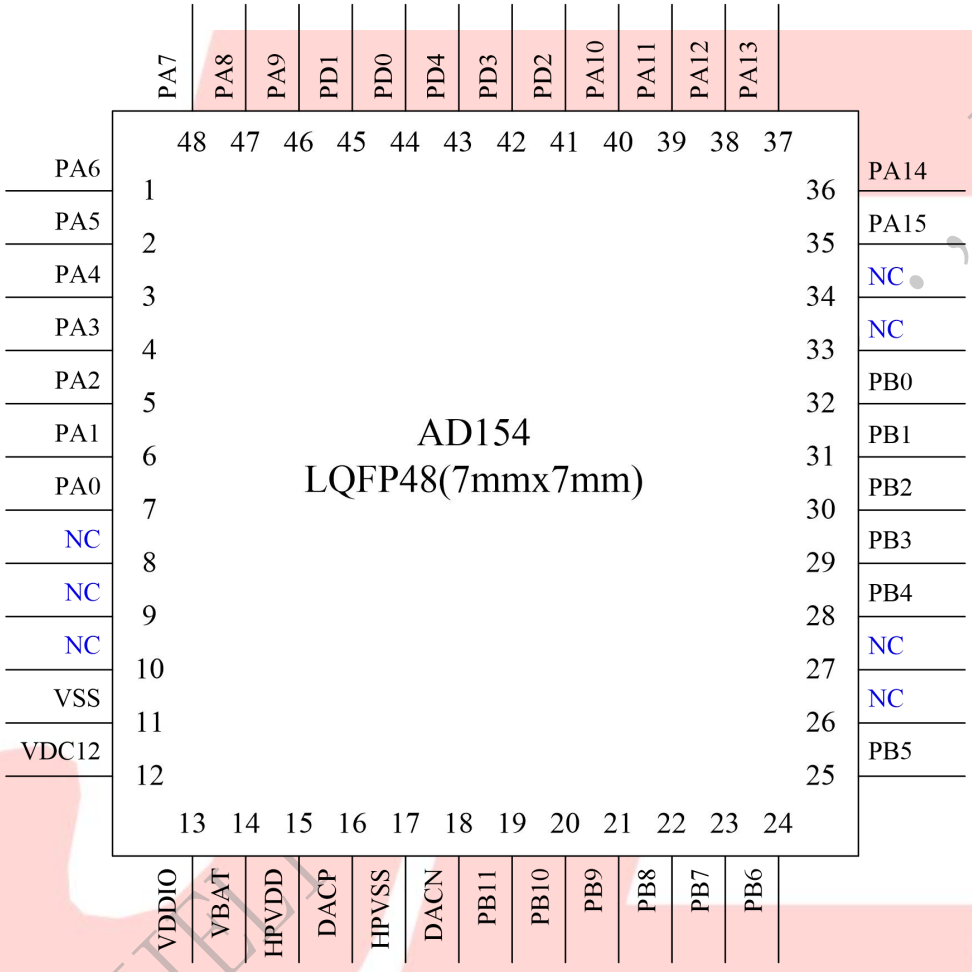


Figure 2-1 AD154_LQFP48 Package Diagram

ZHUHAI JIELI LTD

Confidential
The information contained herein is the exclusive property of JIELI and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of JIELI.

2.2 Pin Description

Table 2-1 AD154_LQFP48 Pin Description

PIN NO.	Name	Type	Drive (mA)	Function	Description
1	PA6	I/O	8/64	GPIO	ADC6:ADC Input Channel 6; SPI1CLKC:SPI1 Clock(C); SD0CLKD:SD0 Clock(D); UART0TXA:Uart0 Data Out(A); I2C_SCL(C); TMR2:Timer2 Clock In; PWM0:PWM Channel0 Output;
2	PA5	I/O	8/64	GPIO	ADC5:ADC Input Channel 5; SPI0DAT3:SPI0 Data 3 UART1RXA:Uart1 Data In(A);
3	PA4	I/O	8/64	GPIO	ADC4:ADC Input Channel 4; SPI0DAT2:SPI0 Data 2; UART1TXA:Uart1 Data Out(A); LVD:Low Voltage Detect；
4	PA3	I/O	8/64	GPIO	ADC3:ADC Input Channel 3; SPI0DIB(1):SPI0 Data1 In(B); SD0DATA:SD0 Data(A); CLKOUT; PWM2L; MCAP0:Motor Timer0 Capture;
5	PA2	I/O	8/64	GPIO	ADC2:ADC Input Channel 2; SPI0DOB(0):SPI0 Data0 Out(B); SD0CMDA:SD0 Command(A); I2C_SDA(B); PWM2H;
6	PA1	I/O	8/64	GPIO	ADC1:ADC Input Channel 1; SPI0CLKB:SPI0 Clock(B); SD0CLKA:SD0 Clock(A); UART0RXB:Uart0 Data In(B); I2C_SCL(B); CAP2:Timer2 Capture;
7	PA0	I/O	8/64	GPIO (pull up)	Long Press Reset; ADC0:ADC Input Channel 0; UART0TXB:Uart0 Data Out(B);
8	NC	_	_		_
9	NC	_	_		_
10	NC	_	_		_

Confidential
The information contained herein is the exclusive property of JIELI and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of JIELI.

ZHUHAI JIELI TECHNOLOGY CO., LTD

11	VSS	G	/		Digital Ground;
12	VDC12	P	/		PLL Power;(Connect the bypass capacitor)
13	VDDIO	P	/		Digital Power; (Internal linear regulator output)
14	VBAT	P	/		Battery Power Supply;
15	HPVDD	P	/		Class-D APA Power Supply;
16	DACP	O	/		Class-D APA Positive Output;
17	HPVSS	G	/		Class-D APA Ground;
18	DACN	O	/		Class-D APA Negative Output;
19	PB11	I/O	8	GPIO (High Voltage Resistance)	OSCIB:Crystal Oscillator Input(B);
20	PB10	I/O	8	GPIO (pull up) (High Voltage Resistance)	MCLR(0 effective);
21	PB9	I/O	8	GPIO (High Voltage Resistance)	SPI1DOD:SPI1 Data Out(D); UART1TRXB:Uart1 Data In/Out(B); I2C_SDA(D); CAP1:Timer1 Capture;
22	PB8	I/O	8	GPIO (High Voltage Resistance)	SPI1CLKD:SPI1 Clock(D); I2C_SCL(D); OSCIA:Crystal Oscillator Input(A);
23	PB7	I/O	8/64	GPIO	SPI1DID:SPI1 Data In(D);
24	PB6	I/O	8/64	GPIO	SD0DATC:SD0 Data(C);
25	PB5	I/O	8/64	GPIO	ADC13:ADC Input Channel 13; SD0CMDC:SD0 Command(C);
26	NC	_	_		_
27	NC	_	_		_
28	PB4	I/O	8/64	GPIO	ADC12:ADC Input Channel 12; SD0CLKC:SD0 Clock(C);
29	PB3	I/O	8/64	GPIO	TDM_MCLK;
30	PB2	I/O	8/64	GPIO	SPI1DIA:SPI1 Data In(A); SD0DATB:SD0 Data(B); TDM_DAT;
31	PB1	I/O	8/64	GPIO (pull down)	ADC11:ADC Input Channel 11; SPI1DOA:SPI1 Data Out(A); SD0CMDB:SD0 Command(B); I2C_SDA(A); TDM_SYN;

ZHUHAI JIELI TECHNOLOGY CO., LTD

Confidential
The information contained herein is the exclusive property of JIELI and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of JIELI.

32	PB0	I/O	8/64	GPIO (pull down)	ADC10:ADC Input Channel 10; SPI1CLKA:SPI1 Clock(A); SD0CLKB:SD0 Clock(B); I2C_SCL(A); TDM_CLK;
33	NC	_	_		_
34	NC	_	_		_
35	PA15	I/O	8/64	GPIO	ADC9:ADC Input Channel 9; SPI1DOB:SPI1 Data Out(B); MCAP3:Motor Timer3 Capture;
36	PA14	I/O	8/64	GPIO	ADC8:ADC Input Channel 8; SPI1CLKB:SPI1 Clock(B); CAP0:Timer0 Capture; MCAP2:Motor Timer2 Capture;
37	PA13	I/O	8/64	GPIO	SPI1DIB:SPI1 Data In(B); TMR1:Timer1 Clock In; MCAP1:Motor Timer1 Capture;
38	PA12	I/O	8/64	GPIO	PWM3:PWM Channel3 Output;
39	PA11	I/O	8/64	GPIO	TMR0:Timer0 Clock In; PWM2:PWM Channel2 Output;
40	PA10	I/O	8/64	GPIO	
41	PD2	I/O	8/64	GPIO (pull up)	SPI0CSA:SPI0 Chip Select(A); SFCCSA:SFC Chip Select(A);
42	PD3	I/O	8/64	GPIO	SPI0DIA(1):SPI0 Data1 In(A); SFCDIA(1):SFC Data1 In(A);
43	PD4	I/O	8/64	GPIO	Flash Power Gate;
44	PD0	I/O	8/64	GPIO	SPI0CLKA:SPI0 Clock(A); SFCCLKA:SFC Clock(A);
45	PD1	I/O	8/64	GPIO	SPI0DOA(0):SPI0 Data0 Out(A); SFCDOA(0):SFC Data0 Out(A);
46	PA9	I/O	8/64	GPIO	
47	PA8	I/O	8/64	GPIO	SPI1DIC:SPI1 Data In(C); SD0DATD:SD0 Data(D);
48	PA7	I/O	8/64	GPIO	ADC7:ADC Input Channel 7; SPI1DOC:SPI1 Data Out(C); SD0CMDD:SD0 Command(D); UART0RXA:Uart0 Data In(A); I2C_SDA(C); PWM1:PWM Channel1 Output;

ZHUHAI JIELI TECHNOLOGY CO., LTD

Confidential
The information contained herein is the exclusive property of JIELI and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of JIELI.

3 Electrical Characteristics

3.1 Absolute Maximum Ratings

Table 3-1

Symbol	Parameter	Min	Max	Unit
Tamb	Ambient Temperature	-40	+85	°C
Tstg	Storage temperature	-65	+150	°C
VBAT	Supply Voltage	-0.3	5.5	V
HPVDD	Class D Audio Power Amplifier	-0.3	5.5	V
$V_{VDDIO33}$	3.3V IO Input Voltage	-0.3	3.6	V

Note : The chip can be damaged by any stress in excess of the absolute maximum ratings listed below.

3.2 PMU Characteristics

Table 3-2

Symbol	Parameter	Min	Typ	Max	Unit	Test Conditions
VBAT	Voltage Input	2.0	3.7	5.5	V	_
V_{HPVDD}	Voltage Input	2.0	3.7	5.5	V	_
V_{VDDIO}	Voltage output	2.0	3.0	3.4	V	VBAT = 3.7V, 100mA loading
V_{VDC12}	Voltage output	1.05	1.2	1.4	V	VBAT=3.7V
I_{VDDIO}	Loading current	_	_	100	mA	VBAT=3.7V

3.3 IO Input/Output Electrical Logical Characteristics

Table 3-3

IO input characteristics						
Symbol	**Parameter**	**Min**	**Typ**	**Max**	**Unit**	**Test Conditions**
V_{IL}	Low-Level Input Voltage	-0.3	_	0.3* VDDIO	V	VDDIO = 3.3V
V_{IH}	High-Level Input Voltage	0.7* VDDIO	_	VDDIO+0.3	V	VDDIO = 3.3V
IO output characteristics						
V_{OL}	Low-Level Output Voltage	_	_	0.33	V	VDDIO = 3.3V
V_{OH}	High-Level Output Voltage	2.7	_	_	V	VDDIO = 3.3V

ZHUHAI JIELI TECHNOLOGY CO., LTD

Confidential
The information contained herein is the exclusive property of JIELI and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of JIELI.

3.4 Internal Resistor Characteristics

Table 3-4

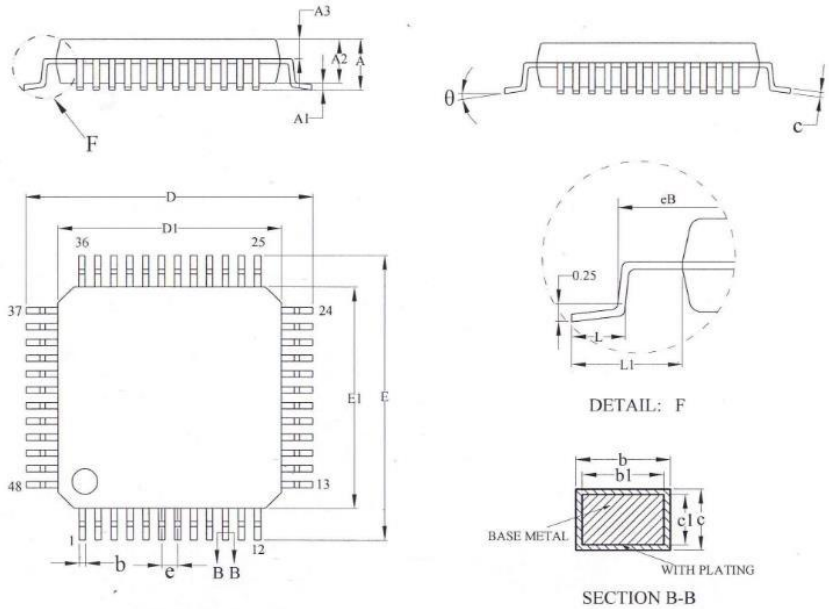
Port	General Output	High Drive	Internal Pull-Up Resistor	Internal Pull-Down Resistor	Comment
PA0~PA15 PB0~PB7 PD0~PD4	8mA	64mA	10K	60K	1、PA0,PB10,PD2 default pull up 2、PB0 & PB1 default pull down 3、internal pull-up/pull-down resistance | accuracy ±20%
PB8~PB11	8mA	_	10K	60K	

ZHUHAI JIELI TECHNOLOGY CO., LTD

Confidential
The information contained herein is the exclusive property of JIELI and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of JIELI.

4 Package Information

4.1 LQFP48(7mm*7mm)



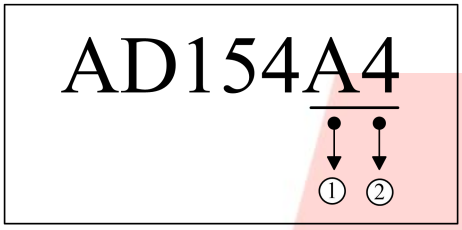
SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A	—	—	1.60
A1	0.05	—	0.15
A2	1.35	1.40	1.45
A3	0.59	0.64	0.69
b	0.18	—	0.26
b1	0.17	0.20	0.23
c	0.13	—	0.17
c1	0.12	0.13	0.14
D	8.80	9.00	9.20
D1	6.90	7.00	7.10
E	8.80	9.00	9.20
E1	6.90	7.00	7.10
eB	8.10	—	8.25
e	0.50BSC		
L	0.40	—	0.65
L1	1.00REF		
θ	0	—	7°

Figure 4-1. AD154_LQFP48 Package

ZHUHAI JIELI TECHNOLOGY

Confidential
The information contained herein is the exclusive property of JIELI and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of JIELI.

5 Package Type Specification



①Represents different packages

②Represents different memory sizes

0: No memory

2: 2Mbit Flash

4: 4Mbit Flash

ZHUHAI JIELI TECHNOLOGY CO., LTD

Confidential

The information contained herein is the exclusive property of JIELI and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of JIELI.

6 Storage Condition

6.1 Moisture Sensitivity Level

AD154A is qualified to moisture sensitivity level MSL3 in accordance with JEDEC J-STD-033.

6.2 Storage Alert

1. Calculated shelf life in sealed bag 12 months at＜40°C and 90% relative humidity (RH).
2. Peak package body temperature≤260°C.
3. After bag is opened,devices that will be subjected to reflow solder or other high temperature process must be mounted within 168 hours of factory conditions≤30°C/60%RH or stored per J-STD-033.
4. Devices require bake before mounting if humidity indicator card reads＞10% for level 2a-5a devices or＞60% for level 2 devices when read at 23±5°C,or 3a or 3b are not met.

Please refer to IPC/JEDEC J-STD-033 for baking procedure if necessary.

ZHUHAI JIELI TECHNOLOGY CO., LTD

Confidential
The information contained herein is the exclusive property of JIELI and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of JIELI.

7 Revision History

Date	Revision	Description
2021.03.23	V1.0	Initial Release
2023.03.21	V1.1	Modify the Features.
2024.11.20	V1.2	Add Block Diagram Add Storage Condition Update Features modification.

ZHUHAI JIELI TECHNOLOGY CO., LTD

Confidential
The information contained herein is the exclusive property of JIELI and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of JIELI.